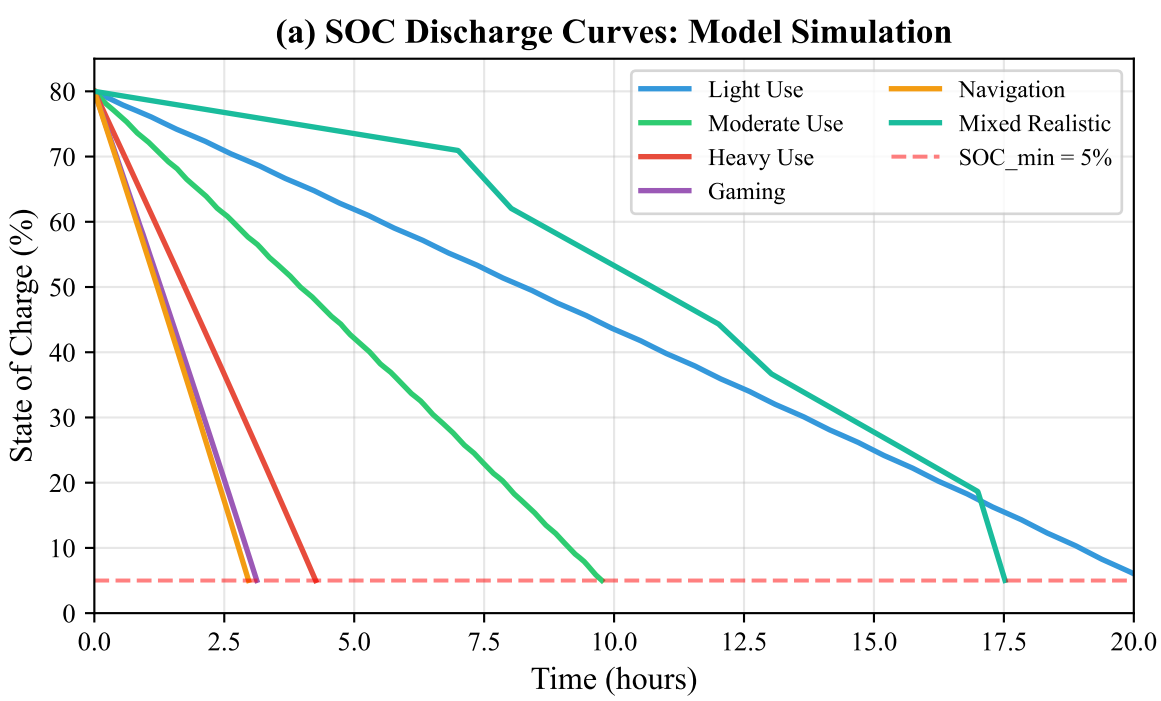
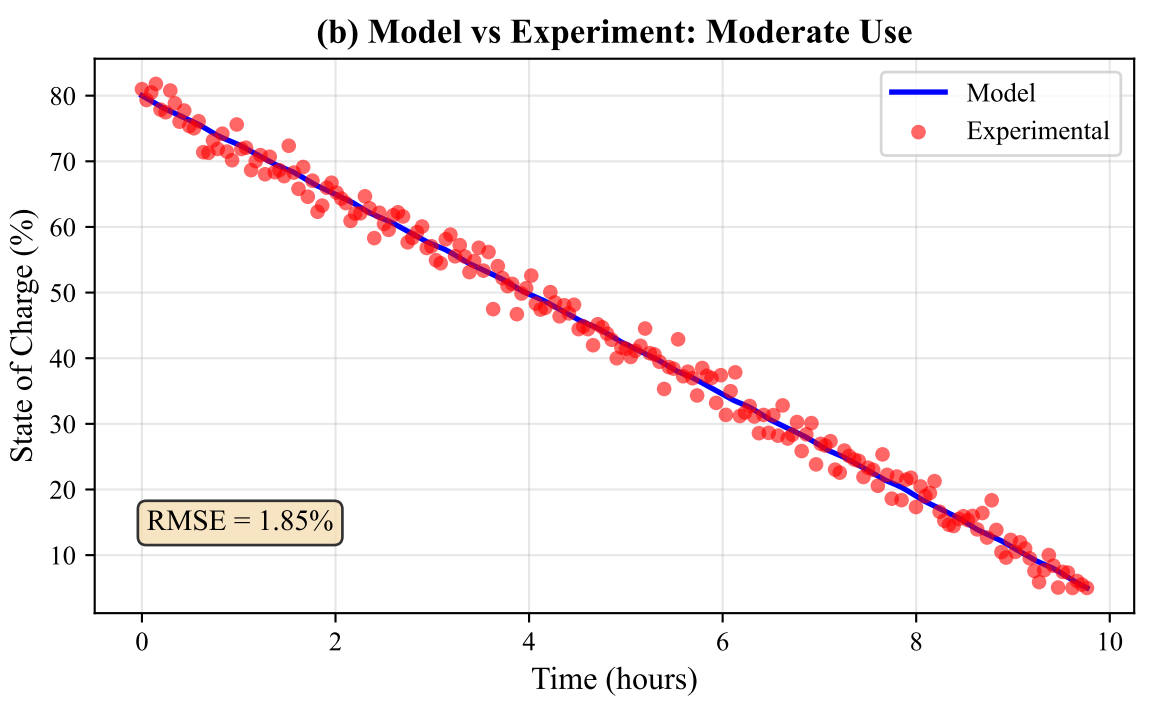
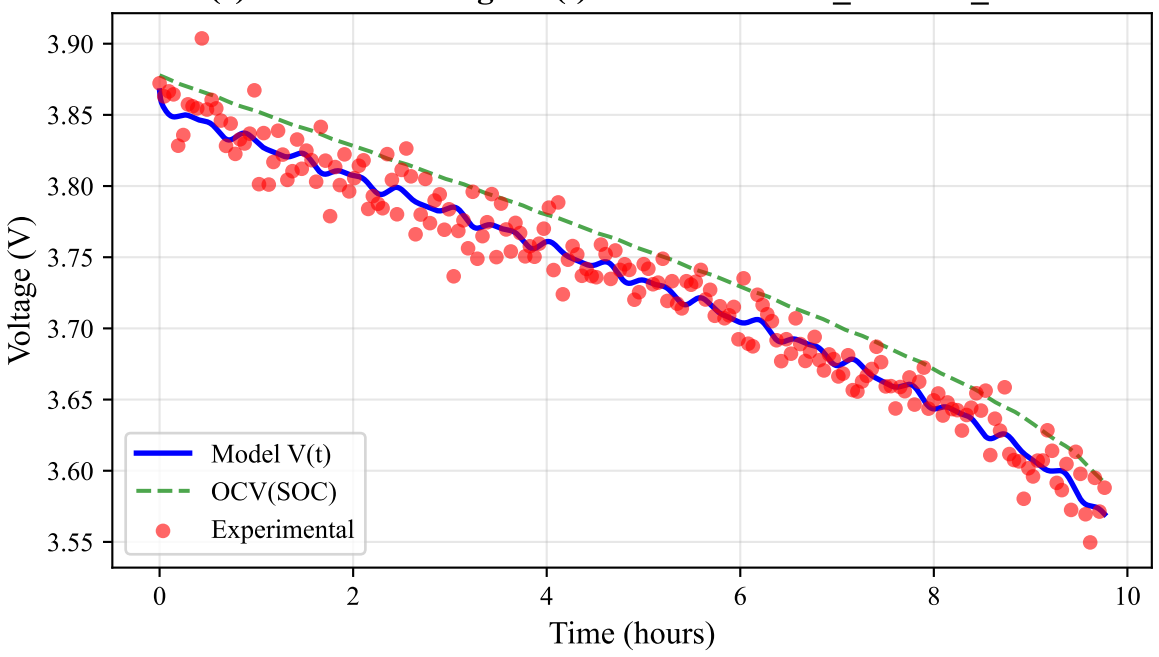
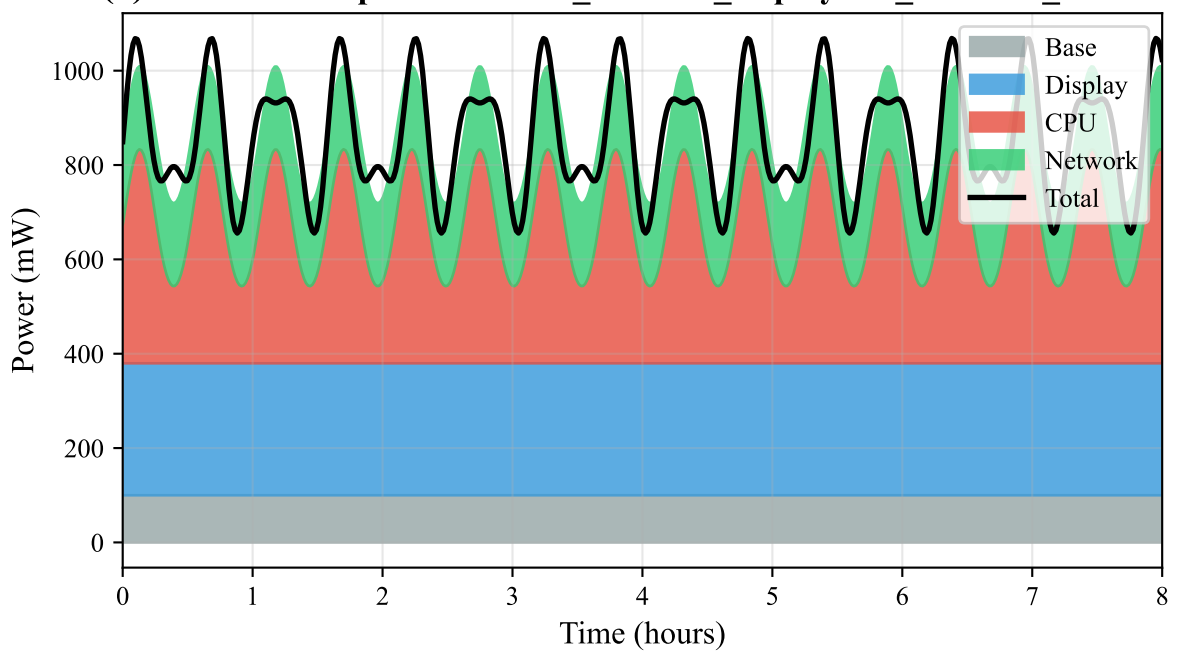
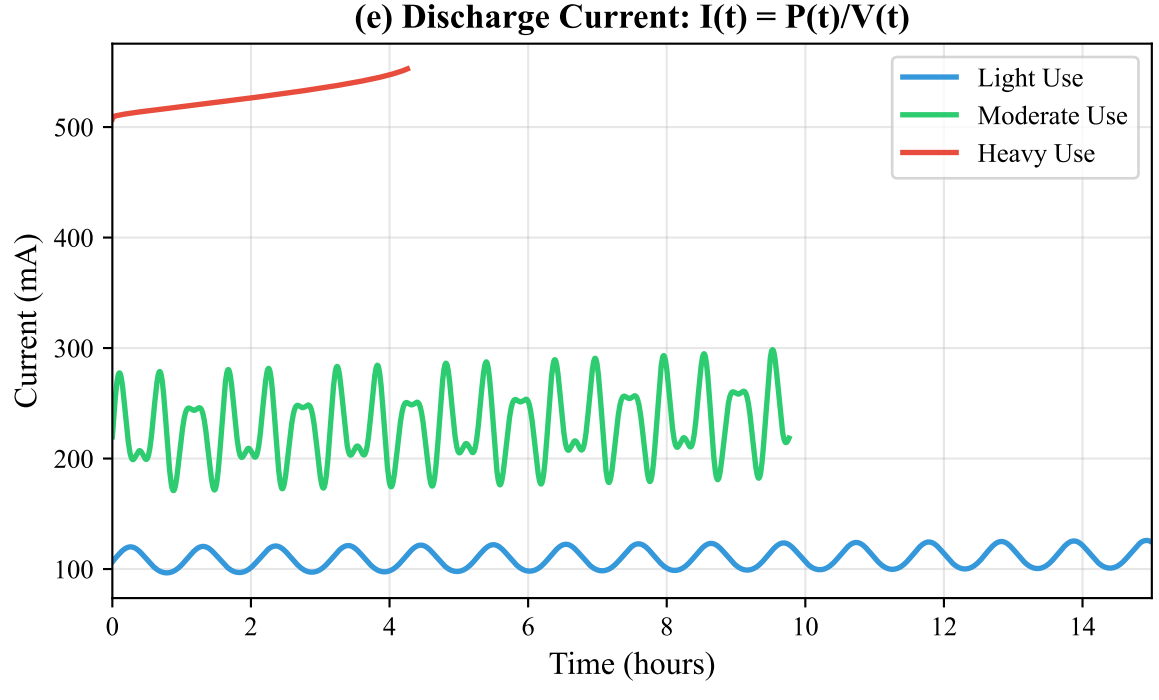
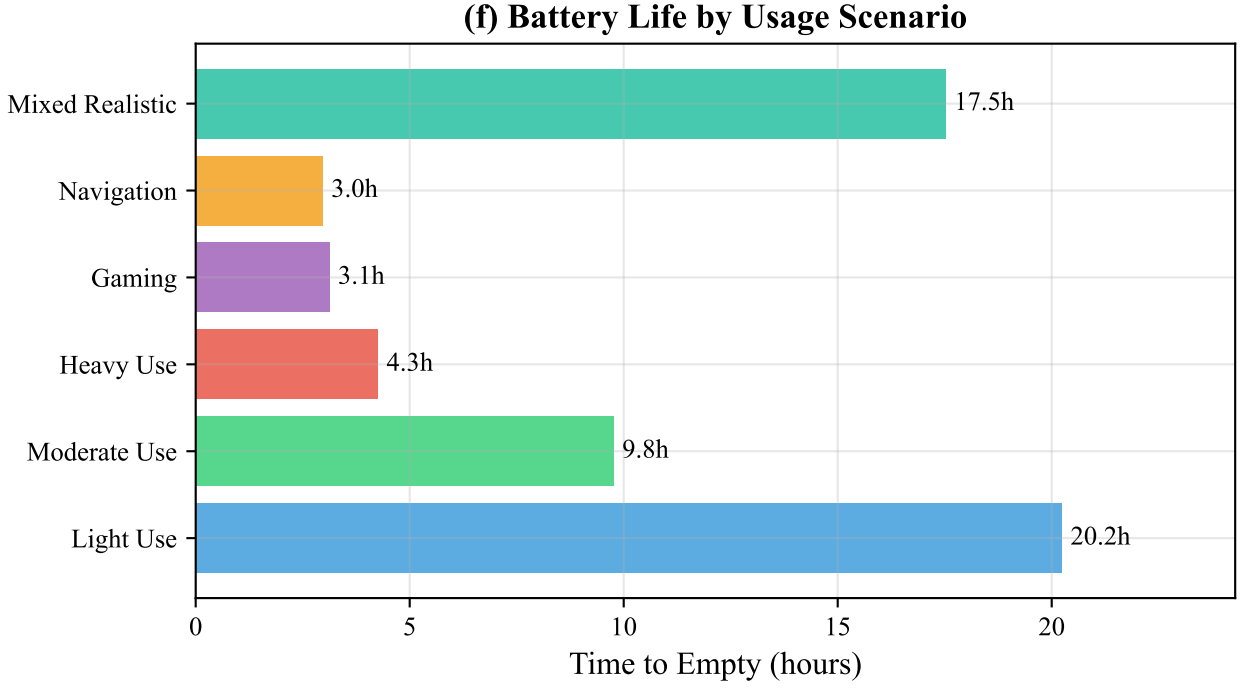

(a) SOC Discharge Curves: Model Simulation
Light Use
Moderate Use
Heavy Use
Gaming
Navigation
Mixed Realistic
SOC_min = 5%
State of Charge (%)
Time (hours)


(b) Model vs Experiment: Moderate Use
Model
Experimental
RMSE = 1.85%
State of Charge (%)
Time (hours)


(c) Terminal Voltage: V(t) = OCV - IR₀ - V_RC1 - V_RC2
Model V(t)
OCV(SOC)
Experimental
Voltage (V)
Time (hours)


(d) Power Decomposition: P = P_base + P_display + P_CPU + P_network
Base
Display
CPU
Network
Total
Power (mW)
Time (hours)


(e) Discharge Current: I(t) = P(t)/V(t)
Light Use
Moderate Use
Heavy Use
Current (mA)
Time (hours)


(f) Battery Life by Usage Scenario
Mixed Realistic
17.5h
Navigation
3.0h
Gaming
3.1h
Heavy Use
4.3h
Moderate Use
9.8h
Light Use
20.2h
Time to Empty (hours)